RET fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes RET gene fusions from the msk_impact_50k_2026 study, covering 194 fusion events. 146/171 fusions with known frame status (85.4%) were in-frame. 179/194 fusions (92.3%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.00256613, statistically significant at alpha=0.05). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 85.4% in-frame and 92.3% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 194 analyzed fusion events, identifying 33 distinct fusion partner genes. The most frequent partner was KIF5B, observed in 87 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 141/146 (96.6%) of in-frame fusions retained the domain, p=0.00256613 (statistically significant at alpha=0.05). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 9.9999e-06.

Domain disruption

Disruption of the Cadherin domain was tested with Fisher's exact test comparing in-frame fusions against all others; 111/116 (95.7%) of in-frame fusions disrupted the domain, p=0.00534494 (statistically significant at alpha=0.05). The Cadherin domain appears to require loss or disruption rather than retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 9.9999e-06.

Junction concentration

90% of 143 fusions with this gene as the 3' partner join at exon 12, against 4% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=9.9999e-06, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 157 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 712 aa (exon 11; label-permutation p=0.000999001). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -12 aa from the nearest configured domain boundary at 724 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=145) and "not_retained" (n=12). For Frame_status, retained: 112/128 (87.5%, 95% CI 80.7%-92.2%); not_retained: 4/11 (36.4%, 95% CI 15.2%-64.6%). Welch's t-test comparing Tumor_variant_count between groups gave means of 61.4276 vs 36.75, p=0.0186911 (statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score produced results for this run (33 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for RET in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (10 row(s) in genomic_bin_recurrence, 2 row(s) in exact_position_recurrence, 5 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: RET has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run (4 row(s) in retention_counter_intuitive_events, 2 row(s) in retention_counter_intuitive_recurrent_partners, 5 row(s) in disruption_counter_intuitive_events, 2 row(s) in disruption_counter_intuitive_recurrent_partners); see the accompanying table.

Mutation cooccurrence

Mutation cooccurrence reported: RET has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (93 row(s) in window_scan, 51 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIF5B	87	0.4484536082474227	0.5000004342923104	0.5000004342923104	0.9841784989858012	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.576686355805218	1
CCDC6	51	0.26288659793814434	0.5000004342923104	0.5000004342923104	0.9805370424031682	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.5204700342250396	2
NCOA4	16	0.08247422680412371	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.46896150688946975	3
SPECC1L	4	0.020618556701030927	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.45040480585854187	4
ERCC6	3	0.015463917525773196	0.5000004342923104	0.5000004342923104	1.0	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.449162673740853	5
TIMM23B	3	0.015463917525773196	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.44885841410596455	6
RUFY3	2	0.010309278350515464	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.44731202235338724	7
SLC4A4	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	1.0	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4460698902356984	8
CCDC186	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4457656306008099	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CLIP1	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4457656306008099	10
GEMIN5	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4457656306008099	11
GRIPAP1	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4457656306008099	12
KIF13A	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4457656306008099	13
RELCH	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4457656306008099	14
RUFY2	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4457656306008099	15
SHROOM3	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4457656306008099	16
TRIM24	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.9979716024340771	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4457656306008099	17
EML4	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.920892494929006	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.43420376447504927	18
RUFY1	2	0.010309278350515464	0.5000004342923104	0.5000004342923104	0.9006085192697769	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4327075598787422	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ERC1	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.8519269776876268	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4238589368888424	20
PDCD10	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.8275862068965517	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.42020782127018114	21
LIN52	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.7464503042596349	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4080374358746436	22
DOCK1	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.6835699797160244	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.398605387193102	23
KCNMA1	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.6713995943204868	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3967798293837714	24
CTNNA3	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.5496957403651115	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3785242512904651	25
KIAA1217	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.41784989858012167	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3587473750227166	26
CTNNA1	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.3772819472616633	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35266218232494784	27
CUBN	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	None	0.6952330305158133	confidence_certainty, domain_disruption, domain_retention, recurrence	0.348317517924351	28
RASSF4	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	None	0.6952330305158133	confidence_certainty, domain_disruption, domain_retention, recurrence	0.348317517924351	29
RBPMS	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	None	0.6952330305158133	confidence_certainty, domain_disruption, domain_retention, recurrence	0.348317517924351	30

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TFG	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	None	0.6952330305158133	confidence_certainty, domain_disruption, domain_retention, recurrence	0.348317517924351	31
FBXL7	2	0.010309278350515464	0.5000004342923104	0.5000004342923104	0.28803245436105473	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.34082115014243386	32
UXS1	1	0.005154639175257732	0.5000004342923104	0.5000004342923104	0.0	0.6952330305158133	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.29606989023569835	33

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
219	0	145	1	11	0.0	0.07643312101910828	1.116718406361609
425	1	144	1	11	0.0763888888888889	0.14747672709456153	0.8312765091222271
550	3	142	2	10	0.1056338028169014	0.04724149779081552	1.3256763415417077
556	4	141	2	10	0.14184397163120568	0.06782347643241221	1.1686199533264545
587	5	140	3	9	0.10714285714285714	0.015547976772642942	1.808326116839365
622	6	139	3	9	0.12949640287769784	0.02228965194918904	1.6518967129341526
627	7	138	5	7	0.07101449275362319	0.0006374545945137115	3.195550744312713
639	8	137	5	7	0.08175182481751825	0.0009948438290156487	3.0022450896261272
640	9	136	5	7	0.09264705882352942	0.001485978263157586	2.8279875433747432
663	10	135	5	7	0.1037037037037037	0.0021399083069381804	2.669604835362925
671	11	134	5	7	0.11492537313432835	0.002987672928356682	2.5246669480831163
673	12	133	5	7	0.12631578947368421	0.004061908949823156	2.3912698157989034
687	13	132	5	7	0.13787878787878788	0.005396440622644964	2.2678925971449133
708	14	131	5	7	0.14961832061068703	0.007025858533551756	2.153300599023978
712	14	131	11	1	0.009715475364330326	1.9508147586058598e-09	8.709783967476186
713	143	2	11	1	6.5	0.21347852886788568	0.6706457986863291
934	143	2	12	0	0.0	1.0	-0.0
1019	144	1	12	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

111	33
5	8

frame_domain_contingency_tables_by_tumor_type

stratum	table
BRCANOS	[1, 0], [0, 0]
COAD	[4, 4], [0, 1]
CSCLC	[1, 0], [0, 0]
CUP	[1, 1], [0, 0]
ESCA	[0, 1], [0, 0]
GBC	[0, 1], [0, 0]
ILC	[2, 0], [0, 0]
LNET	[1, 0], [0, 0]
LUAD	[77, 11], [3, 2]
LUNE	[1, 0], [0, 0]
LUPC	[1, 0], [0, 0]
NSCLCPD	[1, 0], [0, 0]
PHC	[0, 0], [0, 1]
PRAD	[0, 0], [0, 2]
SARCNOS	[1, 0], [0, 0]
STAD	[1, 2], [0, 0]
THAP	[1, 0], [0, 0]
THPA	[14, 11], [1, 1]
THPD	[4, 2], [1, 1]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
cadherin	Cadherin domain	157	13	1	143	0.003885582010582011

domain_retention tables

frame_domain_contingency_table

112	33

4	8

frame_domain_contingency_tables_by_tumor_type

stratum	table
BRCANOS	[1, 0], [0, 0]
COAD	[4, 4], [0, 1]
CSCLC	[1, 0], [0, 0]
CUP	[1, 1], [0, 0]
ESCA	[0, 1], [0, 0]
GBC	[0, 0], [0, 1]
ILC	[2, 0], [0, 0]
LNET	[1, 0], [0, 0]
LUAD	[78, 11], [2, 2]
LUNE	[1, 0], [0, 0]
LUPC	[1, 0], [0, 0]
NSCLCPD	[1, 0], [0, 0]
PHC	[0, 1], [0, 0]
PRAD	[0, 0], [0, 2]
SARCNOS	[1, 0], [0, 0]
STAD	[1, 2], [0, 0]
THAP	[1, 0], [0, 0]
THPA	[14, 11], [1, 1]
THPD	[4, 2], [1, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	157	145	1	11	0.06235224586288416

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
12	146	157	0.9299363057324841

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
CCDC186	1	1	1	1
CCDC6	51	51	51	42
CLIP1	1	1	1	1
CTNNA1	1	1	1	1
CTNNA3	1	1	1	1
CUBN	1	1	1	1
DOCK1	1	1	1	1
EML4	1	1	1	1
ERC1	1	1	1	1
ERCC6	3	3	3	1
FBXL7	2	2	2	1
GEMIN5	1	1	1	1
GRIPAP1	1	1	1	1
KCNMA1	1	1	1	1
KIAA1217	1	1	1	1
KIF13A	1	1	1	1
KIF5B	87	87	86	71
LIN52	1	1	1	1
NCOA4	16	16	16	16
PDCD10	1	1	1	1
RASSF4	1	1	1	1
RBPMS	1	1	1	1
RELCH	1	1	1	1
RUFY1	2	2	2	1
RUFY2	1	1	1	1
RUFY3	2	2	2	1
SHROOM3	1	1	1	1
SLC4A4	1	1	1	1
SPECC1L	4	4	4	2
TFG	1	1	1	1
TIMM23B	3	3	3	3
TRIM24	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
UXS1	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
10	43611000	43612000	81	EVT-P-0000208-T01-IM3-70, EVT-P-0000889-T01-IM3-154, EVT-P-0001151-T01-IM3-179, EVT-P-0001745-T02-IM5-151, EVT-P-0001898-T01-IM3-153, ... (81 total)
10	43610000	43611000	58	EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0001010-T02-IM5-53, EVT-P-0001155-T01-IM3-180, EVT-P-0004094-T02-IM6-131, ... (58 total)
10	43609000	43610000	6	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0038256-T01-IM6-28, EVT-P-0040111-T01-IM6-14, EVT-P-0049588-T01-IM6-152, ... (6 total)
10	43608000	43609000	5	EVT-P-0005168-T01-IM5-148, EVT-P-0019663-T01-IM6-118, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, EVT-P-0030396-T02-IM6-164
10	43622000	43623000	2	EVT-P-0045809-T02-IM6-170, EVT-P-0063502-T01-IM7-162
10	43600000	43601000	1	EVT-P-0046790-T01-IM6-223
10	43606000	43607000	1	EVT-P-0026245-T01-IM6-30
10	43607000	43608000	1	EVT-P-0018069-T01-IM6-45
10	43612000	43613000	1	EVT-P-0036737-T02-IM6-76
10	43617000	43618000	1	EVT-P-0063802-T01-IM7-163

exact_position_recurrence

chromosome	position	n_events	event_ids
10	43610353	3	EVT-P-0004916-T01-IM5-144, EVT-P-0031330-T01-IM6-141, EVT-P-0036400-T03-IM6-229
10	43610308	2	EVT-P-0015588-T01-IM6-132, EVT-P-0055183-T01-IM6-104

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
713	129	126	1814	False
712	6	6	1663	False
550	3	3	369	False
627	3	3	117	False
587	2	2	184	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	12	129	143	0.9020979020979021	0.03741209563994374	165.3626654494909
three_prime	11	8	143	0.055944055944055944	0.019896858884200657	2.087175750850472
five_prime	11	6	14	0.42857142857142855	0.03945616502578528	5.06497514469768
three_prime	9	3	143	0.02097902097902098	0.013858415377402719	0.4971007512492097
five_prime	10	2	14	0.14285714285714285	0.017327707454289732	1.6235363947646926
five_prime	19	2	14	0.14285714285714285	0.028823253633380216	1.221197262481129
three_prime	10	2	143	0.013986013986013986	0.013352086263478669	0.24364462395528072
five_prime	4	1	14	0.07142857142857142	0.026704172526957337	0.5011191388633525
five_prime	8	1	14	0.07142857142857142	0.014139709329582747	0.7429161963968159
five_prime	9	1	14	0.07142857142857142	0.013183309892170652	0.7706890623167427
five_prime	16	1	14	0.07142857142857142	0.03234880450070324	0.4330324066541603
three_prime	7	1	143	0.006993006993006993	0.04191279887482419	0.0009531940777002335

mechanistic_interpretation tables

retention_counter_intuitive_events

event_id	sample_id	partner_gene	breakpoint_protein_position	domain_status
EVT-P-0003125-T01-IM5-81	P-0003125-T01-IM5	KIF5B	712	lost
EVT-P-0000425-T01-IM3-147	P-0000425-T01-IM3	CCDC6	712	lost
EVT-P-0000889-T01-IM3-154	P-0000889-T01-IM3	CCDC6	712	lost
EVT-P-0043704-T01-IM6-174	P-0043704-T01-IM6	KIF5B	712	lost

retention_counter_intuitive_recurrent_partners

partner_gene	count
CCDC6	2
KIF5B	2

disruption_counter_intuitive_events

event_id	sample_id	partner_gene	breakpoint_protein_position	domain_status
EVT-P-0003125-T01-IM5-81	P-0003125-T01-IM5	KIF5B	712	retained
EVT-P-0000425-T01-IM3-147	P-0000425-T01-IM3	CCDC6	712	retained
EVT-P-0000889-T01-IM3-154	P-0000889-T01-IM3	CCDC6	712	retained
EVT-P-0045809-T02-IM6-170	P-0045809-T02-IM6	FBXL7	1063	retained
EVT-P-0043704-T01-IM6-174	P-0043704-T01-IM6	KIF5B	712	retained

disruption_counter_intuitive_recurrent_partners

partner_gene	count
CCDC6	2
KIF5B	2

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
350	550	200	4	153	3	142	0.2323943661971831	0.2747659162288297	0.5610371410149504	EVT-P-0018069-T01-IM6-45, EVT-P-0026245-T01-IM6-30, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66
356	556	200	5	152	4	141	0.3120567375886525	0.33164702083833325	0.4793238995846121	EVT-P-0018069-T01-IM6-45, EVT-P-0026245-T01-IM6-30, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, EVT-P-0030396-T02-IM6-164
387	587	200	7	150	5	140	0.17857142857142858	0.09088619687054571	1.0415020692328363	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0026245-T01-IM6-30, EVT-P-0027417-T01-IM6-173, ... (7 total)
422	622	200	8	149	6	139	0.2158273381294964	0.11599893690317994	0.9355459909385844	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0026245-T01-IM6-30, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
425	625	200	8	149	6	139	0.2158273381294964	0.11599893690317994	0.9355459909385844	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0026245-T01-IM6-30, ... (8 total)
427	627	200	10	147	6	139	0.08633093525179857	0.0032945266206773805	2.4822069788839114	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (10 total)
439	639	200	11	146	7	138	0.10144927536231885	0.004955907844745455	2.3048767779397665	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (11 total)
440	640	200	12	145	8	137	0.11678832116788321	0.007115134469861328	2.147816887713653	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (12 total)
456	556	100	4	153	3	142	0.2323943661971831	0.2747659162288297	0.5610371410149504	EVT-P-0018069-T01-IM6-45, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, EVT-P-0030396-T02-IM6-164
463	663	200	13	144	9	136	0.1323529411764706	0.00983526364357052	2.0072139939131013	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (13 total)
471	671	200	14	143	10	135	0.14814814814814814	0.013174977795735695	1.8802501080469216	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (14 total)
473	673	200	15	142	11	134	0.16417910447761194	0.017187730337116607	1.7647814686965901	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0013886-T01-IM5-165, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, ... (15 total)
487	587	100	6	151	4	141	0.14184397163120568	0.06782347643241221	1.1686199533264545	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
487	687	200	16	141	12	133	0.18045112781954886	0.021921082806703247	1.6591379973883225	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0013886-T01-IM5-165, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, ... (16 total)
506	556	50	4	153	3	142	0.2323943661971831	0.2747659162288297	0.5610371410149504	EVT-P-0018069-T01-IM6-45, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, EVT-P-0030396-T02-IM6-164
508	708	200	17	140	13	132	0.19696969696969696	0.027416213224204823	1.5619925309251597	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0013886-T01-IM5-165, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, ... (17 total)
512	712	200	23	134	13	132	0.019696969696969695	3.410541810970232e-08	7.467176621939211	EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0002145-T01-IM3-183, EVT-P-0003125-T01-IM5-81, EVT-P-0005168-T01-IM5-148, ... (23 total)
513	713	200	152	5	142	3	9.466666666666667	0.047241497790815525	1.3256763415417074	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (152 total)
522	622	100	7	150	5	140	0.17857142857142858	0.09088619687054571	1.0415020692328363	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0027417-T01-IM6-173, ... (7 total)
527	627	100	10	147	6	139	0.08633093525179857	0.0032945266206773805	2.4822069788839114	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (10 total)
531	556	25	4	153	3	142	0.2323943661971831	0.2747659162288297	0.5610371410149504	EVT-P-0018069-T01-IM6-45, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, EVT-P-0030396-T02-IM6-164
537	587	50	6	151	4	141	0.14184397163120568	0.06782347643241221	1.1686199533264545	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
539	639	100	11	146	7	138	0.10144927536231885	0.004955907844745455	2.3048767779397665	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (11 total)
540	640	100	12	145	8	137	0.11678832116788321	0.007115134469861328	2.147816887713653	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (12 total)
550	575	25	4	153	3	142	0.2323943661971831	0.2747659162288297	0.5610371410149504	EVT-P-0018069-T01-IM6-45, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, EVT-P-0030396-T02-IM6-164
550	600	50	6	151	4	141	0.14184397163120568	0.06782347643241221	1.1686199533264545	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, ... (6 total)
550	650	100	12	145	8	137	0.11678832116788321	0.007115134469861328	2.147816887713653	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (12 total)
550	750	200	152	5	142	3	9.466666666666667	0.047241497790815525	1.3256763415417074	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (152 total)
556	656	100	9	148	6	139	0.12949640287769784	0.02228965194918904	1.6518967129341526	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0030396-T02-IM6-164, ... (9 total)
556	756	200	149	8	140	5	9.333333333333334	0.015547976772642944	1.8083261168393647	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (149 total)
563	663	100	9	148	6	139	0.12949640287769784	0.02228965194918904	1.6518967129341526	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0029222-T02-IM6-212, ... (9 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
571	671	100	10	147	7	138	0.15217391304347827	0.030430419947122627	1.5166920542320532	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0029222-T02-IM6-212, ... (10 total)
573	673	100	11	146	8	137	0.17518248175182483	0.039983361985514054	1.3981206911991513	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0013886-T01-IM5-165, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (11 total)
577	627	50	6	151	3	142	0.06338028169014084	0.006077540507622167	2.216272137844042	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0049588-T01-IM6-152, ... (6 total)
587	637	50	6	151	3	142	0.06338028169014084	0.006077540507622167	2.216272137844042	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0049588-T01-IM6-152, ... (6 total)
587	687	100	12	145	9	136	0.19852941176470587	0.05093943782406497	1.2929458522906148	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0013886-T01-IM5-165, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (12 total)
587	787	200	148	9	139	6	7.722222222222222	0.02228965194918905	1.6518967129341524	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (148 total)
589	639	50	5	152	3	142	0.1056338028169014	0.04724149779081552	1.3256763415417077	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0038256-T01-IM6-28, EVT-P-0049588-T01-IM6-152, EVT-P-0064853-T01-IM7-23
590	640	50	6	151	4	141	0.14184397163120568	0.06782347643241221	1.1686199533264545	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0038256-T01-IM6-28, EVT-P-0040111-T01-IM6-14, EVT-P-0049588-T01-IM6-152, ... (6 total)
602	627	25	4	153	2	143	0.06993006993006994	0.029616366785053857	1.5284682200208528	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0049588-T01-IM6-152, EVT-P-0064853-T01-IM7-23

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
608	708	100	11	146	9	136	0.33088235294117646	0.1999420692283573	0.6990958176303357	EVT-P-0002145-T01-IM3-183, EVT-P-0013886-T01-IM5-165, EVT-P-0024225-T01-IM6-55, EVT-P-0027723-T01-IM6-65, EVT-P-0029222-T02-IM6-212, ... (11 total)
612	712	100	17	140	9	136	0.03308823529411765	1.2611589611023586e-06	5.899230169906984	EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0002145-T01-IM3-183, EVT-P-0003125-T01-IM5-81, EVT-P-0013886-T01-IM5-165, ... (17 total)
613	663	50	7	150	5	140	0.17857142857142858	0.09088619687054571	1.0415020692328363	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0040111-T01-IM6-14, ... (7 total)
613	713	100	146	11	138	7	9.857142857142858	0.004955907844745455	2.3048767779397665	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (146 total)
614	639	25	5	152	3	142	0.1056338028169014	0.04724149779081552	1.3256763415417077	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0038256-T01-IM6-28, EVT-P-0049588-T01-IM6-152, EVT-P-0064853-T01-IM7-23
615	640	25	6	151	4	141	0.14184397163120568	0.06782347643241221	1.1686199533264545	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0038256-T01-IM6-28, EVT-P-0040111-T01-IM6-14, EVT-P-0049588-T01-IM6-152, ... (6 total)
621	671	50	8	149	6	139	0.2158273381294964	0.11599893690317994	0.9355459909385844	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0039257-T01-IM6-84, ... (8 total)
622	647	25	6	151	4	141	0.14184397163120568	0.06782347643241221	1.1686199533264545	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0038256-T01-IM6-28, EVT-P-0040111-T01-IM6-14, EVT-P-0049588-T01-IM6-152, ... (6 total)
622	672	50	8	149	6	139	0.2158273381294964	0.11599893690317994	0.9355459909385844	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0039257-T01-IM6-84, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
622	722	100	146	11	138	7	9.857142857142858	0.004955907844745455	2.3048767779397665	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (146 total)
622	822	200	146	11	138	7	9.857142857142858	0.004955907844745455	2.3048767779397665	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (146 total)
623	673	50	8	149	6	139	0.2158273381294964	0.11599893690317994	0.9355459909385844	EVT-P-0002145-T01-IM3-183, EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0039257-T01-IM6-84, ... (8 total)
627	652	25	5	152	3	142	0.1056338028169014	0.04724149779081552	1.3256763415417077	EVT-P-0002145-T01-IM3-183, EVT-P-0038256-T01-IM6-28, EVT-P-0040111-T01-IM6-14, EVT-P-0049588-T01-IM6-152, EVT-P-0064853-T01-IM7-23
627	677	50	8	149	6	139	0.2158273381294964	0.11599893690317994	0.9355459909385844	EVT-P-0002145-T01-IM3-183, EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0039257-T01-IM6-84, ... (8 total)
627	727	100	145	12	137	8	8.5625	0.007115134469861328	2.147816887713653	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (145 total)
627	827	200	145	12	137	8	8.5625	0.007115134469861328	2.147816887713653	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (145 total)
637	687	50	6	151	6	139	None	1.0	-0.0	EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0039257-T01-IM6-84, EVT-P-0040111-T01-IM6-14, ... (6 total)
639	689	50	6	151	6	139	None	1.0	-0.0	EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0039257-T01-IM6-84, EVT-P-0040111-T01-IM6-14, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
639	739	100	142	15	136	9	15.11111111111111	0.00017811817559639924	3.7492917618479913	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (142 total)
639	839	200	142	15	136	9	15.11111111111111	0.00017811817559639924	3.7492917618479913	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (142 total)
640	690	50	5	152	5	140	None	1.0	-0.0	EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0039257-T01-IM6-84, EVT-P-0040111-T01-IM6-14, EVT-P-0059913-T01-IM7-109
640	740	100	141	16	135	10	13.5	0.0002748261398174729	3.560941962156213	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (141 total)
640	840	200	141	16	135	10	13.5	0.0002748261398174729	3.560941962156213	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (141 total)
658	708	50	5	152	5	140	None	1.0	-0.0	EVT-P-0013886-T01-IM5-165, EVT-P-0027723-T01-IM6-65, EVT-P-0029222-T02-IM6-212, EVT-P-0039257-T01-IM6-84, EVT-P-0059913-T01-IM7-109
662	687	25	4	153	4	141	None	1.0	-0.0	EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0039257-T01-IM6-84, EVT-P-0059913-T01-IM7-109
662	712	50	11	146	5	140	0.03571428571428571	1.897608502373887e-05	4.721793382423235	EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0003125-T01-IM5-81, EVT-P-0013886-T01-IM5-165, EVT-P-0014808-T01-IM6-166, ... (11 total)
663	688	25	4	153	4	141	None	1.0	-0.0	EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0039257-T01-IM6-84, EVT-P-0059913-T01-IM7-109
663	713	50	140	17	134	11	12.181818181818182	0.0004095064768371502	3.387739224959997	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (140 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
663	763	100	140	17	134	11	12.181818181818182	0.0004095064768371502	3.387739224959997	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (140 total)
663	863	200	140	17	134	11	12.181818181818182	0.0004095064768371502	3.387739224959997	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (140 total)
671	721	50	139	18	133	12	11.083333333333334	0.0005921266004864504	3.2275854283877035	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (139 total)
671	771	100	139	18	133	12	11.083333333333334	0.0005921266004864504	3.2275854283877035	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (139 total)
671	871	200	139	18	133	12	11.083333333333334	0.0005921266004864504	3.2275854283877035	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (139 total)
673	723	50	138	19	132	13	10.153846153846153	0.0008340704721059438	3.0787972534944674	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (138 total)
673	773	100	138	19	132	13	10.153846153846153	0.0008340704721059438	3.0787972534944674	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (138 total)
673	873	200	138	19	132	13	10.153846153846153	0.0008340704721059438	3.0787972534944674	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (138 total)
687	712	25	8	149	2	143	0.013986013986013986	1.2781292111203256e-06	5.893425239417067	EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0003125-T01-IM5-81, EVT-P-0014808-T01-IM6-166, EVT-P-0027723-T01-IM6-65, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
687	737	50	137	20	131	14	9.357142857142858	0.0011481466781213111	2.940002626372254	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (137 total)
687	787	100	137	20	131	14	9.357142857142858	0.0011481466781213111	2.940002626372254	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (137 total)
687	887	200	137	20	131	14	9.357142857142858	0.0011481466781213111	2.940002626372254	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (137 total)
688	713	25	136	21	130	15	8.666666666666666	0.001548574736398811	2.810067830137764	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (136 total)
708	733	25	136	21	130	15	8.666666666666666	0.001548574736398811	2.810067830137764	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (136 total)
708	758	50	136	21	130	15	8.666666666666666	0.001548574736398811	2.810067830137764	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (136 total)
708	808	100	136	21	130	15	8.666666666666666	0.001548574736398811	2.810067830137764	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (136 total)
708	908	200	136	21	130	15	8.666666666666666	0.001548574736398811	2.810067830137764	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (136 total)
712	737	25	135	22	129	16	8.0625	0.0020509500117193873	2.6880449246628353	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (135 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
712	762	50	135	22	129	16	8.0625	0.0020509500117193873	2.6880449246628353	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (135 total)
712	812	100	135	22	129	16	8.0625	0.0020509500117193873	2.6880449246628353	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (135 total)
712	912	200	135	22	129	16	8.0625	0.0020509500117193873	2.6880449246628353	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (135 total)
713	738	25	129	28	129	16	None	9.998230834156442e-11	10.00007684069374	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0001010-T02-IM5-53, EVT-P-0001151-T01-IM3-179, EVT-P-0001155-T01-IM3-180, ... (129 total)
713	763	50	129	28	129	16	None	9.998230834156442e-11	10.00007684069374	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0001010-T02-IM5-53, EVT-P-0001151-T01-IM3-179, EVT-P-0001155-T01-IM3-180, ... (129 total)
713	813	100	129	28	129	16	None	9.998230834156442e-11	10.00007684069374	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0001010-T02-IM5-53, EVT-P-0001151-T01-IM3-179, EVT-P-0001155-T01-IM3-180, ... (129 total)
713	913	200	129	28	129	16	None	9.998230834156442e-11	10.00007684069374	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0001010-T02-IM5-53, EVT-P-0001151-T01-IM3-179, EVT-P-0001155-T01-IM3-180, ... (129 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
713	738	25	129	28	129	16	None	9.998230834156442e-11	10.00007684069374	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0001010-T02-IM5-53, EVT-P-0001151-T01-IM3-179, EVT-P-0001155-T01-IM3-180, ... (129 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
512	712	200	23	134	13	132	0.019696969696969695	3.410541810970232e-08	7.467176621939211	EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0002145-T01-IM3-183, EVT-P-0003125-T01-IM5-81, EVT-P-0005168-T01-IM5-148, ... (23 total)
612	712	100	17	140	9	136	0.03308823529411765	1.2611589611023586e-06	5.899230169906984	EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0002145-T01-IM3-183, EVT-P-0003125-T01-IM5-81, EVT-P-0013886-T01-IM5-165, ... (17 total)
687	712	25	8	149	2	143	0.013986013986013986	1.2781292111203256e-06	5.893425239417067	EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0003125-T01-IM5-81, EVT-P-0014808-T01-IM6-166, EVT-P-0027723-T01-IM6-65, ... (8 total)
662	712	50	11	146	5	140	0.03571428571428571	1.897608502373887e-05	4.721793382423235	EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0003125-T01-IM5-81, EVT-P-0013886-T01-IM5-165, EVT-P-0014808-T01-IM6-166, ... (11 total)
639	739	100	142	15	136	9	15.11111111111111	0.00017811817559639924	3.7492917618479913	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (142 total)
640	740	100	141	16	135	10	13.5	0.0002748261398174729	3.560941962156213	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (141 total)
663	713	50	140	17	134	11	12.181818181818182	0.0004095064768371502	3.387739224959997	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (140 total)
671	721	50	139	18	133	12	11.083333333333334	0.0005921266004864504	3.2275854283877035	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (139 total)
673	723	50	138	19	132	13	10.153846153846153	0.0008340704721059438	3.0787972534944674	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (138 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
687	737	50	137	20	131	14	9.357142857142858	0.0011481466781213111	2.940002626372254	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (137 total)
688	713	25	136	21	130	15	8.666666666666666	0.001548574736398811	2.810067830137764	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (136 total)
712	737	25	135	22	129	16	8.0625	0.0020509500117193873	2.6880449246628353	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (135 total)
527	627	100	10	147	6	139	0.08633093525179857	0.0032945266206773805	2.4822069788839114	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (10 total)
539	639	100	11	146	7	138	0.10144927536231885	0.004955907844745455	2.3048767779397665	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (11 total)
613	713	100	146	11	138	7	9.857142857142858	0.004955907844745455	2.3048767779397665	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (146 total)
577	627	50	6	151	3	142	0.06338028169014084	0.006077540507622167	2.216272137844042	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0049588-T01-IM6-152, ... (6 total)
540	640	100	12	145	8	137	0.11678832116788321	0.007115134469861328	2.147816887713653	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (12 total)
627	727	100	145	12	137	8	8.5625	0.007115134469861328	2.147816887713653	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (145 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
463	663	200	13	144	9	136	0.1323529411764706	0.00983526364357052	2.0072139939131013	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (13 total)
471	671	200	14	143	10	135	0.14814814814814814	0.013174977795735695	1.8802501080469216	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (14 total)
556	756	200	149	8	140	5	9.333333333333334	0.015547976772642944	1.8083261168393647	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (149 total)
473	673	200	15	142	11	134	0.16417910447761194	0.017187730337116607	1.7647814686965901	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0013886-T01-IM5-165, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, ... (15 total)
487	687	200	16	141	12	133	0.18045112781954886	0.021921082806703247	1.6591379973883225	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0013886-T01-IM5-165, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, ... (16 total)
556	656	100	9	148	6	139	0.12949640287769784	0.02228965194918904	1.6518967129341526	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0030396-T02-IM6-164, ... (9 total)
563	663	100	9	148	6	139	0.12949640287769784	0.02228965194918904	1.6518967129341526	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0029222-T02-IM6-212, ... (9 total)
587	787	200	148	9	139	6	7.722222222222222	0.02228965194918905	1.6518967129341524	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (148 total)
508	708	200	17	140	13	132	0.19696969696969696	0.027416213224204823	1.5619925309251597	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0013886-T01-IM5-165, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, ... (17 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
602	627	25	4	153	2	143	0.06993006993006994	0.029616366785053857	1.5284682200208528	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0049588-T01-IM6-152, EVT-P-0064853-T01-IM7-23
571	671	100	10	147	7	138	0.15217391304347827	0.030430419947122627	1.5166920542320532	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0029222-T02-IM6-212, ... (10 total)
573	673	100	11	146	8	137	0.17518248175182483	0.039983361985514054	1.3981206911991513	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0013886-T01-IM5-165, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (11 total)
614	639	25	5	152	3	142	0.1056338028169014	0.04724149779081552	1.3256763415417077	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0038256-T01-IM6-28, EVT-P-0049588-T01-IM6-152, EVT-P-0064853-T01-IM7-23
627	652	25	5	152	3	142	0.1056338028169014	0.04724149779081552	1.3256763415417077	EVT-P-0002145-T01-IM3-183, EVT-P-0038256-T01-IM6-28, EVT-P-0040111-T01-IM6-14, EVT-P-0049588-T01-IM6-152, EVT-P-0064853-T01-IM7-23
513	713	200	152	5	142	3	9.466666666666667	0.047241497790815525	1.3256763415417074	EVT-P-0000208-T01-IM3-70, EVT-P-0000252-T01-IM3-89, EVT-P-0000425-T01-IM3-147, EVT-P-0000889-T01-IM3-154, EVT-P-0001010-T02-IM5-53, ... (152 total)
587	687	100	12	145	9	136	0.19852941176470587	0.05093943782406497	1.2929458522906148	EVT-P-0002145-T01-IM3-183, EVT-P-0005168-T01-IM5-148, EVT-P-0013886-T01-IM5-165, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, ... (12 total)
615	640	25	6	151	4	141	0.14184397163120568	0.06782347643241221	1.1686199533264545	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0038256-T01-IM6-28, EVT-P-0040111-T01-IM6-14, EVT-P-0049588-T01-IM6-152, ... (6 total)
537	587	50	6	151	4	141	0.14184397163120568	0.06782347643241221	1.1686199533264545	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
613	663	50	7	150	5	140	0.17857142857142858	0.09088619687054571	1.0415020692328363	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0040111-T01-IM6-14, ... (7 total)
522	622	100	7	150	5	140	0.17857142857142858	0.09088619687054571	1.0415020692328363	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0027417-T01-IM6-173, ... (7 total)
387	587	200	7	150	5	140	0.17857142857142858	0.09088619687054571	1.0415020692328363	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0026245-T01-IM6-30, EVT-P-0027417-T01-IM6-173, ... (7 total)
621	671	50	8	149	6	139	0.2158273381294964	0.11599893690317994	0.9355459909385844	EVT-P-0002145-T01-IM3-183, EVT-P-0024225-T01-IM6-55, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0039257-T01-IM6-84, ... (8 total)
623	673	50	8	149	6	139	0.2158273381294964	0.11599893690317994	0.9355459909385844	EVT-P-0002145-T01-IM3-183, EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0039257-T01-IM6-84, ... (8 total)
422	622	200	8	149	6	139	0.2158273381294964	0.11599893690317994	0.9355459909385844	EVT-P-0005168-T01-IM5-148, EVT-P-0018069-T01-IM6-45, EVT-P-0019663-T01-IM6-118, EVT-P-0024225-T01-IM6-55, EVT-P-0026245-T01-IM6-30, ... (8 total)
608	708	100	11	146	9	136	0.33088235294117646	0.1999420692283573	0.6990958176303357	EVT-P-0002145-T01-IM3-183, EVT-P-0013886-T01-IM5-165, EVT-P-0024225-T01-IM6-55, EVT-P-0027723-T01-IM6-65, EVT-P-0029222-T02-IM6-212, ... (11 total)
531	556	25	4	153	3	142	0.2323943661971831	0.2747659162288297	0.5610371410149504	EVT-P-0018069-T01-IM6-45, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, EVT-P-0030396-T02-IM6-164
350	550	200	4	153	3	142	0.2323943661971831	0.2747659162288297	0.5610371410149504	EVT-P-0018069-T01-IM6-45, EVT-P-0026245-T01-IM6-30, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66
356	556	200	5	152	4	141	0.3120567375886525	0.33164702083833325	0.4793238995846121	EVT-P-0018069-T01-IM6-45, EVT-P-0026245-T01-IM6-30, EVT-P-0027417-T01-IM6-173, EVT-P-0027991-T01-IM6-66, EVT-P-0030396-T02-IM6-164

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
662	687	25	4	153	4	141	None	1.0	-0.0	EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0039257-T01-IM6-84, EVT-P-0059913-T01-IM7-109
637	687	50	6	151	6	139	None	1.0	-0.0	EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0038256-T01-IM6-28, EVT-P-0039257-T01-IM6-84, EVT-P-0040111-T01-IM6-14, ... (6 total)
640	690	50	5	152	5	140	None	1.0	-0.0	EVT-P-0013886-T01-IM5-165, EVT-P-0029222-T02-IM6-212, EVT-P-0039257-T01-IM6-84, EVT-P-0040111-T01-IM6-14, EVT-P-0059913-T01-IM7-109
658	708	50	5	152	5	140	None	1.0	-0.0	EVT-P-0013886-T01-IM5-165, EVT-P-0027723-T01-IM6-65, EVT-P-0029222-T02-IM6-212, EVT-P-0039257-T01-IM6-84, EVT-P-0059913-T01-IM7-109

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022642-T03-IM6-3	P-0022642-T03-IM6	P-0022642	Thyroid Cancer	THAP	CCDC6-RET	CCDC6	in-frame	three_prime	43611768	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0017365-T04-IM6-4	P-0017365-T04-IM6	P-0017365	Non-Small Cell Lung Cancer	LUAD	CCDC6-RET	CCDC6	in-frame	three_prime	43611464	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068500-T01-IM7-5	P-0068500-T01-IM7	P-0068500	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43610322	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022380-T03-IM6-6	P-0022380-T03-IM6	P-0022380	Non-Small Cell Lung Cancer	LUAD	CCDC6-RET	CCDC6	in-frame	three_prime	43610659	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0004901-T01-IM5-7	P-0004901-T01-IM5	P-0004901	Thyroid Cancer	THPD	CCDC6-RET	CCDC6	in-frame	three_prime	43611729	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : : CCDC6_:RETinv | Protein fusion: in frame (CCDC6-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022642-T04-IM6-8	P-0022642-T04-IM6	P-0022642	Thyroid Cancer	THPD	CCDC6-RET	CCDC6	in-frame	three_prime	43611768	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0036379-T01-IM6-9	P-0036379-T01-IM6	P-0036379	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43611418	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0037412-T01-IM6-10	P-0037412-T01-IM6	P-0037412	Non-Small Cell Lung Cancer	LUAD	CCDC6-RET	CCDC6	in-frame	three_prime	43610839	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057735-T01-IM6-11	P-0057735-T01-IM6	P-0057735	Colorectal Cancer	COAD	CCDC6-RET	CCDC6	in-frame	three_prime	43611887	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+125:CCDC6_c.2137-145:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0038257-T01-IM6-12	P-0038257-T01-IM6	P-0038257	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43610594	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6(NM_005436) - RET (NM_020975.4) Fusion (CCDC6 exon1 fused withRET exon12): c.304-2131:CCDC6_c.2136+410:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0039548-T01-IM6-13	P-0039548-T01-IM6	P-0039548	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43610843	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040111-T01-IM6-14	P-0040111-T01-IM6	P-0040111	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	unknown	three_prime	43609967	11	640	False	640	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv | Protein Fusion: mid-exon {CCDC6:RET}	NA
EVT-P-0041192-T01-IM6-15	P-0041192-T01-IM6	P-0041192	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43610571	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0051781-T03-IM6-16	P-0051781-T03-IM6	P-0051781	Non-Small Cell Lung Cancer	LUAD	CCDC6-RET	CCDC6	in-frame	three_prime	43610377	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042417-T01-IM6-17	P-0042417-T01-IM6	P-0042417	Non-Small Cell Lung Cancer	LUAD	CCDC6-RET	CCDC6	in-frame	three_prime	43610397	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0047144-T01-IM6-18	P-0047144-T01-IM6	P-0047144	Breast Cancer	BRCANOS	CCDC6-RET	CCDC6	in-frame	three_prime	43611888	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8847:CCDC6_c.2137-144:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0048611-T01-IM6-19	P-0048611-T01-IM6	P-0048611	Thyroid Cancer	THPD	CCDC6-RET	CCDC6	in-frame	three_prime	43611237	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049461-T02-IM6-20	P-0049461-T02-IM6	P-0049461	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43611705	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-3450:CCDC6_c.2137-327:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0051781-T01-IM6-21	P-0051781-T01-IM6	P-0051781	Non-Small Cell Lung Cancer	LUAD	CCDC6-RET	CCDC6	in-frame	three_prime	43610377	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0059857-T02-IM7-22	P-0059857-T02-IM7	P-0059857	Non-Small Cell Lung Cancer	LUAD	CCDC6-RET	CCDC6	in-frame	three_prime	43611615	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.453+8003:CCDC6_c.2137-417:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064853-T01-IM7-23	P-0064853-T01-IM7	P-0064853	Colorectal Cancer	COAD	CCDC6-RET	CCDC6	out-of-frame	three_prime	43609511	11	627	True	627	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv | Protein Fusion: out of frame {CCDC6:RET}	NA
EVT-P-0065992-T01-IM7-24	P-0065992-T01-IM7	P-0065992	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43611698	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12221:CCDC6_c.2137-334:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0038053-T01-IM6-26	P-0038053-T01-IM6	P-0038053	Non-Small Cell Lung Cancer	LUAD	CLIP1-RET	CLIP1	unknown	three_prime	43611121	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CLIP1 (NM_001247997) - RET (NM_020975) rearrangement: t(10;12)(q11.21;q24.31)(chr10:g.43611121::chr12:g.122825512) | Protein Fusion: mid-exon {CLIP1:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000252-T01-IM3-27	P-0000252-T01-IM3	P-0000252	Non-Small Cell Lung Cancer	LUAD	CUBN-RET	CUBN	unknown	three_prime	43610494	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA | Antisense fusion	NA
EVT-P-0038256-T01-IM6-28	P-0038256-T01-IM6	P-0038256	Non-Small Cell Lung Cancer	LUAD	ERC1-RET	ERC1	unknown	three_prime	43609964	11	639	False	639	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		ERC1 (NM_178040) - RET (NM_020975) Rearrangement : t(10;12)(q11.1; p13.32)(chr10:g.43609964::chr12:g.126 4106) | Protein Fusion: mid-exon {ERC1:RET}	NA
EVT-P-0034781-T01-IM6-29	P-0034781-T01-IM6	P-0034781	Non-Small Cell Lung Cancer	LUAD	GRIPAP1-RET	GRIPAP1	in-frame	three_prime	43611853	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		GRIPAP1 (NM_020137) - RET (NM_020975) rearrangement: t(X;10)(p11.23q11.21)(chrX:g.48831008::chr10:g.43611853) | Protein Fusion: in frame {GRIPAP1:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026245-T01-IM6-30	P-0026245-T01-IM6	P-0026245	Esophagogastric Cancer	STAD	KIAA1217-RET	KIAA1217	unknown	three_prime	43606666	7	425	False	425	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIAA1217 (NM_019590) - RET (NM_020975) fusion (KIAA1217 exons 1-13 fused to RET exons 7-20): c.2390:KIAA1217_c.1275:RETdel | Protein Fusion: mid-exon {KIAA1217:RET}	NA
EVT-P-0042153-T01-IM6-31	P-0042153-T01-IM6	P-0042153	Non-Small Cell Lung Cancer	LUAD	KIF13A-RET	KIF13A	in-frame	three_prime	43610510	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF13A (NM_022113) - RET (NM_020975) fusion: t(6;10)(p22.3;q11.21)(chr6:g.17806558::chr10:g.43610510) | Protein Fusion: in frame {KIF13A:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032686-T01-IM6-32	P-0032686-T01-IM6	P-0032686	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611982	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2299:KIF5B_c.2137-50:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009514-T01-IM5-33	P-0009514-T01-IM5	P-0009514	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611811	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		Note: The KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozantinib. PMID: 23533264 | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011035-T01-IM5-34	P-0011035-T01-IM5	P-0011035	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611322	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1726-2296:KIF5B_c.2137-710:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0012648-T01-IM5-35	P-0012648-T01-IM5	P-0012648	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611697	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-335:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013825-T01-IM5-36	P-0013825-T01-IM5	P-0013825	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	unknown	three_prime	43611520	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1782:KIF5B_c.2137-512:RETinv | Protein fusion: mid-exon (KIF5B-RET)	NA
EVT-P-0014808-T01-IM6-37	P-0014808-T01-IM6	P-0014808	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611055	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014808-T02-IM6-38	P-0014808-T02-IM6	P-0014808	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611056	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0014808-T03-IM6-39	P-0014808-T03-IM6	P-0014808	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611056	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016221-T01-IM6-40	P-0016221-T01-IM6	P-0016221	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611268	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0016221-T02-IM6-41	P-0016221-T02-IM6	P-0016221	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611268	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017214-T01-IM6-42	P-0017214-T01-IM6	P-0017214	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610647	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused in-frame with RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RET inv | Protein Fusion: in frame {KIF5B: RET}	NA
EVT-P-0017214-T02-IM6-43	P-0017214-T02-IM6	P-0017214	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610647	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused to RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RET inv | Protein Fusion: in frame {KIF5B: RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017523-T02-IM6-44	P-0017523-T02-IM6	P-001752 3	Non-Small Cell Lung Cancer	LUPC	KIF5B-RET	KIF5B	in-frame	three_prime	436111 92	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2362:KIF5B_c.2137-840:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0018069-T01-IM6-45	P-0018069-T01-IM6	P-001806 9	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	out-of-frame	five_prime	436078 32	8	550	True	549	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.1648+160:RET_c.1915-72:KIF5Binv | Protein Fusion: out of frame {RET:KIF5B}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019535-T02-IM6-46	P-0019535-T02-IM6	P-0019535	Non-Small Cell Lung Cancer	LUNE	KIF5B-RET	KIF5B	in-frame	three_prime	43611913	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-2113:KIF5B_c.2137-119:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0020273-T02-IM6-47	P-0020273-T02-IM6	P-0020273	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611819	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021150-T01-IM6-48	P-0021150-T01-IM6	P-0021150	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611591	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021578-T01-IM6-49	P-0021578-T01-IM6	P-0021578	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610809	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021879-T01-IM6-50	P-0021879-T01-IM6	P-0021879	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610699	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B_c.2136+515_RETinv | Protein Fusion: in frame {KIF5B: RET}	NA
EVT-P-0021937-T01-IM6-51	P-0021937-T01-IM6	P-0021937	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611555	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:KIF5B_c.2137-477:RETinv | Protein Fusion: in frame {KIF5B: RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021937-T03-IM6-52	P-0021937-T03-IM6	P-0021937	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611555	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:KIF5B_c.2137-477:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0001010-T02-IM5-53	P-0001010-T02-IM5	P-0001010	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610863	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-780:KIF5B_c.2136+679:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022866-T01-IM6-54	P-0022866-T01-IM6	P-0022866	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611871	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-767:KIF5B_c.2137-161:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0024225-T01-IM6-55	P-0024225-T01-IM6	P-0024225	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	unknown	three_prime	43609110	10	622	False	622	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 10-20): c.2762-430:KIF5B_c.1866:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024843-T01-IM6-56	P-0024843-T01-IM6	P-0024843	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611733	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20): c.1726-1339:KIF5B_c.2137-299:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025100-T01-IM6-57	P-0025100-T01-IM6	P-0025100	Non-Small Cell Lung Cancer	NSCLCPD	KIF5B-RET	KIF5B	in-frame	three_prime	43611511	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2273:KIF5B_c.2137-521:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025100-T02-IM6-58	P-0025100-T02-IM6	P-0025100	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611511	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) and RET (NM_020975) Fusion (KIF5B exon 16 fused to RET exon 12): c.1726-2273:KIF5B_c.2137-788:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025838-T01-IM6-59	P-0025838-T01-IM6	P-0025838	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610972	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025928-T01-IM6-60	P-0025928-T01-IM6	P-0025928	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025928-T02-IM6-61	P-0025928-T02-IM6	P-0025928	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+483:KIF5B_c.2137-122:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026693-T01-IM6-62	P-0026693-T01-IM6	P-0026693	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610511	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion:c.2544+156:KIF5_c.2136+327:RETinv. | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0026777-T01-IM6-63	P-0026777-T01-IM6	P-0026777	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610717	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026777-T02-IM6-64	P-0026777-T02-IM6	P-0026777	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610717	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0027723-T01-IM6-65	P-0027723-T01-IM6	P-0027723	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	unknown	three_prime	43610171	11	708	False	708	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 11-20):c.1725+966:KIF5B_c.2123:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027991-T01-IM6-66	P-0027991-T01-IM6	P-0027991	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43608201	9	550	True	550	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B_c.1649-100:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0030453-T01-IM6-67	P-0030453-T01-IM6	P-0030453	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610911	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032227-T01-IM6-68	P-0032227-T01-IM6	P-0032227	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611125	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2000:KIF5B_c.2137-907:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032407-T01-IM6-69	P-0032407-T01-IM6	P-0032407	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611730	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+230:KIF5B_c.2137-302:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0000208-T01-IM3-70	P-0000208-T01-IM3	P-0000208	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611647	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RET (NM_020975) Inversion (c.2137-385_KIF5B:c.1726-1971inv) | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032753-T01-IM6-71	P-0032753-T01-IM6	P-0032753	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610925	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1323:KIF5B_c.2136+741:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0033700-T02-IM6-72	P-0033700-T02-IM6	P-0033700	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611670	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1454:KIF5B_c.2137-362:RET | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0034357-T01-IM6-73	P-0034357-T01-IM6	P-0034357	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611187	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-845:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036092-T01-IM6-74	P-0036092-T01-IM6	P-0036092	Non-Small Cell Lung Cancer	LNET	KIF5B-RET	KIF5B	in-frame	three_prime	43610413	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1138:KIF5B_c.2136+229:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0036637-T01-IM6-75	P-0036637-T01-IM6	P-0036637	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611965	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036737-T02-IM6-76	P-0036737-T02-IM6	P-0036737	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43612025	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1487:KIF5B_c.2137-7:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0037367-T01-IM6-77	P-0037367-T01-IM6	P-0037367	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611874	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0002694-T02-IM5-78	P-0002694-T02-IM5	P-0002694	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611682	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037494-T01-IM6-79	P-0037494-T01-IM6	P-0037494	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611755	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0002873-T01-IM3-80	P-0002873-T01-IM3	P-0002873	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611244	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0003125-T01-IM5-81	P-0003125-T01-IM5	P-0003125	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	five_prime	43611581	11	712	True	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		NA | Protein fusion: in frame (RET-KIF5B)	NA
EVT-P-0003125-T01-IM5-82	P-0003125-T01-IM5	P-0003125	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611584	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038961-T01-IM6-83	P-0038961-T01-IM6	P-0038961	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610775	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-293:KIF5B_c.2136+591:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039257-T01-IM6-84	P-0039257-T01-IM6	P-0039257	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	671	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0039257-T03-IM7-85	P-0039257-T03-IM7	P-0039257	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	671	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039474-T01-IM6-86	P-0039474-T01-IM6	P-0039474	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610359	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039474-T02-IM6-87	P-0039474-T02-IM6	P-0039474	Non-Small Cell Lung Cancer	NSCLC	KIF5B-RET	KIF5B	in-frame	three_prime	43610359	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039941-T01-IM6-88	P-0039941-T01-IM6	P-0039941	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611339	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2040:KIF5B_c.2137-693:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000252-T01-IM3-89	P-000002 52-T01-IM3	P-000002 52	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610597	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RET (NM_020975) Inversion (c.2136+413_KIF5B:c.1726-153inv) | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0042104-T01-IM6-90	P-004210 4-T01-IM6	P-004210 4	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610505	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1056:KIF5B_c.2136+321:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0004995-T01-IM5-91	P-000499 5-T01-IM5	P-000499 5	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610660	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B(NM_004521)-RET(NM_020975) Fusion (KIF5B exon 15 fused to RET exon 12) : c.1726-2366:KIF5B_c.2136+476:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042580-T01-IM6-92	P-0042580-T01-IM6	P-0042580	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610944	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1619:KIF5B_c.2136+760:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0042608-T01-IM6-93	P-0042608-T01-IM6	P-0042608	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611802	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1670:KIF5B_c.2137-230:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0043007-T01-IM6-94	P-0043007-T01-IM6	P-0043007	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610813	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043007-T02-IM6-95	P-0043007-T02-IM6	P-0043007	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610813	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0046726-T01-IM6-96	P-0046726-T01-IM6	P-0046726	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610987	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+768:KIF5B_c.2136+803:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0046827-T02-IM6-97	P-0046827-T02-IM6	P-0046827	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610889	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1305:KIF5B_c.2136+705:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048829-T01-IM6-98	P-0048829-T01-IM6	P-0048829	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610616	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+948:KIF5B_c.2136+432:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0048968-T01-IM6-99	P-0048968-T01-IM6	P-0048968	Small Cell Lung Cancer	CSCLC	KIF5B-RET	KIF5B	in-frame	three_prime	43611809	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+211:KIF5B_c.2137-223:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0005573-T01-IM5-100	P-0005573-T01-IM5	P-0005573	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610418	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1726-860:KIF5B_c.2136+234:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052826-T01-IM6-101	P-0052826-T01-IM6	P-0052826	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611182	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2358:KIF5B_c.2137-850:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0052861-T02-IM6-102	P-0052861-T02-IM6	P-0052861	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610260	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-20:KIF5B_c.2136+76:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0053175-T01-IM6-103	P-0053175-T01-IM6	P-0053175	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611162	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-282:KIF5B_c.2137-870:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0055183-T01-IM6-104	P-0055183-T01-IM6	P-0055183	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610308	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0055183-T02-IM6-105	P-0055183-T02-IM6	P-0055183	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610308	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0057072-T02-IM6-106	P-0057072-T02-IM6	P-0057072	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611106	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057072-T03-IM7-107	P-0057072-T03-IM7	P-0057072	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611106	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0008101-T01-IM5-108	P-0008101-T01-IM5	P-0008101	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610455	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059913-T01-IM7-109	P-0059913-T01-IM7	P-0059913	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	unknown	three_prime	43610108	11	687	False	687	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2441:KIF5B_c.2060:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0060748-T02-IM7-110	P-0060748-T02-IM7	P-0060748	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611527	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0060748-T03-IM7-111	P-0060748-T03-IM7	P-0060748	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611527	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0061331-T01-IM7-112	P-0061331-T01-IM7	P-0061331	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610884	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-286:KIF5B_c.2136+700:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0062948-T03-IM7-113	P-0062948-T03-IM7	P-0062948	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43610211	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B_c.2136+27:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008289-T01-IM5-114	P-0008289-T01-IM5	P-0008289	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611748	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20) : c.1726-824:KIF5B_c.2137-284:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0064951-T01-IM7-115	P-0064951-T01-IM7	P-0064951	Cancer of Unknown Primary	CUP	KIF5B-RET	KIF5B	out-of-frame	three_prime	43611130	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv | Protein Fusion: out of frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065876-T01-IM7-116	P-0065876-T01-IM7	P-0065876	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611299	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0000208-T02-IM5-117	P-0000208-T02-IM5	P-0000208	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	KIF5B	in-frame	three_prime	43611647	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019663-T01-IM6-118	P-0019663-T01-IM6	P-0019663	Prostate Cancer	PRAD	LIN52-RET	LIN52	out-of-frame	five_prime	43608613	9	587	True	586	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020975) - LIN52 (NM_001024674) Rearrangement: t(10;14)(p32.1;q12)(chr10:g.43608613:chr14:g.74562750) | Protein Fusion: out of frame {RET:LIN52}	NA
EVT-P-0038315-T01-IM6-120	P-0038315-T01-IM6	P-0038315	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	out-of-frame	three_prime	43611981	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-415:NCOA4_c.2137-51:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049600-T01-IM6-121	P-0049600-T01-IM6	P-0049600	Colorectal Cancer	COAD	NCOA4-RET	NCOA4	unknown	three_prime	43611613	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.1889:NCOA4_c.2137-419:RETdup | Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-0043077-T01-IM6-122	P-0043077-T01-IM6	P-0043077	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	out-of-frame	three_prime	43611580	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0047991-T02-IM6-123	P-0047991-T02-IM6	P-0047991	Thyroid Cancer	THPD	NCOA4-RET	NCOA4	out-of-frame	three_prime	43611538	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+537:NCOA4_c.2137-494:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048805-T01-IM6-124	P-0048805-T01-IM6	P-0048805	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	unknown	three_prime	43611225	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.851:NCOA4_c.2137-807:RETdup | Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-0052409-T01-IM6-125	P-0052409-T01-IM6	P-0052409	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	out-of-frame	three_prime	43611691	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0005748-T01-IM5-126	P-0005748-T01-IM5	P-0005748	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	in-frame	three_prime	43611325	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.763-260:NCOA4_c.2137-707:RETdup | Protein fusion: in frame (NCOA4-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037961-T01-IM6-127	P-0037961-T01-IM6	P-0037961	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	out-of-frame	three_prime	43611448	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.763-360:NCOA4_c.2137-584:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0065826-T01-IM7-128	P-0065826-T01-IM7	P-0065826	Colorectal Cancer	COAD	NCOA4-RET	NCOA4	out-of-frame	three_prime	43610857	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2136+673:RET_c.*23+937NCOA4dup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0056870-T01-IM6-129	P-0056870-T01-IM6	P-0056870	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	out-of-frame	three_prime	43610337	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0060765-T02-IM7-130	P-0060765-T02-IM7	P-0060765	Thyroid Cancer	THPD	NCOA4-RET	NCOA4	out-of-frame	three_prime	43611844	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2137-188:RET_c.763-448NCOA4dup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0004094-T02-IM6-131	P-0004094-T02-IM6	P-0004094	Non-Small Cell Lung Cancer	LUAD	RELCH-RET	RELCH	in-frame	three_prime	43610279	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KIAA1468 (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.21;q21.33)(chr10:g.43610279::chr18:g.59909399) | Protein Fusion: in frame {KIAA1468:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015588-T01-IM6-132	P-0015588-T01-IM6	P-0015588	Non-Small Cell Lung Cancer	LUAD	RET-CCDC186	CCDC186	in-frame	three_prime	43610308	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		C10orf118 (NM_018017) - RET (NM_020975) Rearrangement : c.1326+142:C10orf118_c.2136+124:RETinv | Protein fusion: in frame (C10orf118-RET)	NA
EVT-P-0008955-T03-IM5-133	P-0008955-T03-IM5	P-0008955	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43611019	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv | Protein fusion: in frame (CCDC6-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022380-T01-IM6-134	P-0022380-T01-IM6	P-0022380	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43610659	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:RETinv | Protein Fusion: in frame {CCDC6: RET}	NA
EVT-P-0022380-T02-IM6-135	P-0022380-T02-IM6	P-0022380	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43610659	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET(NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:c.2136+475inv | Protein Fusion: in frame {CCDC6: RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013023-T01-IM5-136	P-0013023-T01-IM5	P-0013023	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43611810	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0030990-T03-IM6-137	P-0030990-T03-IM6	P-0030990	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43611881	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12-20): c.303+10940:CCDC6_c.2137-151:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022607-T01-IM6-138	P-0022607-T01-IM6	P-0022607	Thyroid Cancer	THPA	RET-CCDC6	CCDC6	in-frame	three_prime	43611212	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12 - 20): c.304-414:CCDC6_c.2137-820:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022642-T01-IM6-139	P-0022642-T01-IM6	P-0022642	Thyroid Cancer	THPA	RET-CCDC6	CCDC6	in-frame	three_prime	43611768	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022642-T02-IM6-140	P-0022642-T02-IM6	P-0022642	Thyroid Cancer	THHC	RET-CCDC6	CCDC6	in-frame	three_prime	43611768	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20) : c.304-14214:CCDC6_c.2137-273:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0031330-T01-IM6-141	P-0031330-T01-IM6	P-0031330	Thyroid Cancer	THAP	RET-CCDC6	CCDC6	in-frame	three_prime	43610353	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+3703:CCDC6_c.2136+169:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001339 4-T01-IM5-142	P-0013394-T01-IM5	P-0013394	Thyroid Cancer	THPA	RET-CCDC6	CCDC6	in-frame	three_prime	43610232	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-001736 5-T02-IM6-143	P-0017365-T02-IM6	P-0017365	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43611464	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004916-T01-IM5-144	P-0004916-T01-IM5	P-0004916	Thyroid Cancer	THPA	RET-CCDC6	CCDC6	in-frame	three_prime	43610353	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6(NM_005436)-RET(NM_020975) Fusion(CCDC6 exon 2 fused to RET exon11) : c.304-1153:CCDC6_c.2136+169:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0017365-T03-IM6-145	P-0017365-T03-IM6	P-0017365	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43611464	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025349-T03-IM6-146	P-0025349-T03-IM6	P-0025349	Breast Cancer	ILC	RET-CCDC6	CCDC6	in-frame	three_prime	43611986	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0000425-T01-IM3-147	P-0000425-T01-IM3	P-0000425	Thyroid Cancer	THPA	RET-CCDC6	CCDC6	in-frame	five_prime	43610450	11	712	True	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv) | Protein fusion: in frame (RET-CCDC6)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005168-T01-IM5-148	P-0005168-T01-IM5	P-0005168	Esophagogastric Cancer	STAD	RET-CCDC6	CCDC6	out-of-frame	three_prime	43608797	10	587	True	587	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 Exon1 fused to RET Exon10) : c.304-11493:CCDC6_c.1760-207:RETinv | Protein fusion: out of frame (CCDC6-RET)	NA
EVT-P-0005556-T01-IM5-149	P-0005556-T01-IM5	P-0005556	Thyroid Cancer	THPA	RET-CCDC6	CCDC6	in-frame	three_prime	43611828	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 with RET exons 12-20) - c.303+21888:CCDC6_c.2137-204:RETinv | Protein fusion: in frame (CCDC6-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020974-T01-IM6-150	P-0020974-T01-IM6	P-0020974	Thyroid Cancer	THPD	RET-CCDC6	CCDC6	in-frame	three_prime	43611184	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.303+2543:CCDC6_c.2137-848:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0001745-T02-IM5-151	P-0001745-T02-IM5	P-0001745	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43611245	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - (RET (NM_020975) rearrangement: c.304-9458:CCDC6_c.2137-787:RETinv | Protein fusion: in frame (CCDC6-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049588-T01-IM6-152	P-0049588-T01-IM6	P-0049588	Thyroid Cancer	THPA	RET-CCDC6	CCDC6	out-of-frame	five_prime	43609437	10	627	True	626	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314:RET_c.304-15222:CCDC6inv | Protein Fusion: out of frame {RET:CCDC6}	NA
EVT-P-0001898-T01-IM3-153	P-0001898-T01-IM3	P-0001898	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43611699	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) inversion: c.304-12339_c.2137-333inv5 | Protein fusion: in frame (CCDC6-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000889-T01-IM3-154	P-0000889-T01-IM3	P-0000889	Thyroid Cancer	THPD	RET-CCDC6	CCDC6	in-frame	five_prime	43611994	11	712	True	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Reciprocal Inversion: c.2137-38_c.2137-38inv | Protein fusion: in frame (RET-CCDC6)	NA
EVT-P-0007322-T02-IM6-155	P-0007322-T02-IM6	P-0007322	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43610268	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.454-1512:CCDC6_c.2136+84:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0027953-T01-IM6-156	P-0027953-T01-IM6	P-0027953	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43611278	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+20901:CCDC6_c.2137-754:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015435-T01-IM6-157	P-0015435-T01-IM6	P-0015435	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43610969	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0029315-T01-IM6-158	P-0029315-T01-IM6	P-0029315	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43610631	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20: c.304-20462:CCDC6_c.2136+447:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029315-T02-IM6-159	P-0029315-T02-IM6	P-0029315	Non-Small Cell Lung Cancer	LUAD	RET-CCDC6	CCDC6	in-frame	three_prime	43610631	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-20462:CCDC6_c.2136+447:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0029602-T01-IM6-160	P-0029602-T01-IM6	P-0029602	Thyroid Cancer	THPD	RET-CCDC6	CCDC6	in-frame	three_prime	43611160	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-5768:CCDC6_c.2137-872:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008504-T03-IM6-161	P-0008504-T03-IM6	P-0008504	Colorectal Cancer	COAD	RET-CCDC6	CCDC6	in-frame	three_prime	43610583	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0063502-T01-IM7-162	P-0063502-T01-IM7	P-0063502	Pheochromocytoma	PHC	RET-CTNNA1	CTNNA1	unknown	five_prime	43622038	19	1019	False	1018	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain; Cadherin domain			RET (NM_020975) - CTNNA1 (NM_001903) fusion: t(5;10)(q31.2;q11.21)(chr5:g.138223827::chr10:g.43622038) | Protein Fusion: mid-exon {RET:CTNNA1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0063802-T01-IM7-163	P-0063802-T01-IM7	P-0063802	Prostate Cancer	PRAD	RET-CTNNA3	CTNNA3	out-of-frame	five_prime	43617566	16	934	True	933	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.7482269503541	True	Cadherin domain		Protein kinase domain	RET (NM_020975) - CTNNA3 (NM_013266) rearrangement: c.2801+102:RET_c.1375-34298:CTNNA3inv | Protein Fusion: out of frame {RET:CTNNA3}	NA
EVT-P-0030396-T02-IM6-164	P-0030396-T02-IM6	P-0030396	Non-Small Cell Lung Cancer	LUAD	RET-DOCK1	DOCK1	unknown	three_prime	43608320	9	556	False	556	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		DOCK1 (NM_001380) - RET (NM_020975) Rearrangement: c.2515+295:RET_c.1668:RETdup | Protein Fusion: mid-exon {DOCK1:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013886-T01-IM5-165	P-0013886-T01-IM5	P-0013886	Non-Small Cell Lung Cancer	LUAD	RET-EML4	EML4	unknown	three_prime	43610067	11	673	False	673	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		EML4 (NM_019063) - RET (NM_020975) rearrangement: t(2;10)(p21;q11.21)(chr2:g.42550559::chr10:g.43610067) | Protein fusion: mid-exon (EML4-RET)	NA
EVT-P-0014808-T01-IM6-166	P-0014808-T01-IM6	P-0014808	Non-Small Cell Lung Cancer	LUAD	RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	11	712	True	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv | Protein fusion: out of frame (RET-ERCC6)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014808-T03-IM6-167	P-0014808-T03-IM6	P-0014808	Non-Small Cell Lung Cancer	LUAD	RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	11	712	True	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv | Protein Fusion: out of frame {RET:ERCC6}	NA
EVT-P-0014808-T02-IM6-168	P-0014808-T02-IM6	P-0014808	Non-Small Cell Lung Cancer	LUAD	RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	11	712	True	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv | Protein Fusion: out of frame {RET:ERCC6}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045809-T06-IM6-169	P-0045809-T06-IM6	P-0045809	Non-Small Cell Lung Cancer	LUAD	RET-FBXL7	FBXL7	in-frame	five_prime	43622323	19	1063	True	1062	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain; Cadherin domain			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.155567734::chr10:g.43622323) | Protein Fusion: in frame {RET:FBXL7}	NA
EVT-P-0045809-T02-IM6-170	P-0045809-T02-IM6	P-0045809	Non-Small Cell Lung Cancer	LUAD	RET-FBXL7	FBXL7	in-frame	five_prime	43622269	19	1063	True	1062	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain; Cadherin domain			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.155567734::chr10:g.43622269) | Protein Fusion: in frame {RET:FBXL7}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025215-T01-IM6-171	P-0025215-T01-IM6	P-0025215	Colorectal Cancer	COAD	RET-GEMIN5	GEMIN5	in-frame	three_prime	43611067	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		GEMIN5 (NM_015465) - RET (NM_020975) rearrangement: t(5;10)(q33.2;q11.21)(chr5:g.154274993::chr10:g.43611067) | Protein Fusion: in frame {GEMIN5:RET}	NA
EVT-P-0027417-T01-IM6-173	P-0027417-T01-IM6	P-0027417	Esophagogastric Cancer	ESCA	RET-KCNMA1	KCNMA1	out-of-frame	three_prime	43608158	9	550	True	550	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		KCNMA1 (NM_001161352) - RET (NM_020975) rearrangement: c.378+15627:KCNMA1_c.1649-143:RET | Protein Fusion: out of frame {KCNMA1:RET}	NA
EVT-P-0043704-T01-IM6-174	P-0043704-T01-IM6	P-0043704	Non-Small Cell Lung Cancer	LUAD	RET-KIF5B	KIF5B	in-frame	five_prime	43610331	11	712	True	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.2136+147:RET_c.2545-77:KIF5Binv | Protein Fusion: in frame {RET:KIF5B}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006226-T02-IM5-178	P-0006226-T02-IM5	P-0006226	Cancer of Unknown Primary	CUP	RET-NCOA4	NCOA4	in-frame	three_prime	43611893	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup | Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-0001151-T01-IM3-179	P-0001151-T01-IM3	P-0001151	Colorectal Cancer	COAD	RET-NCOA4	NCOA4	in-frame	three_prime	43611645	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2137-387_c.1840-75dup | Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-0001155-T01-IM3-180	P-0001155-T01-IM3	P-0001155	Non-Small Cell Lung Cancer	LUAD	RET-NCOA4	NCOA4	in-frame	three_prime	43610941	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2136+757_c.715-617dup | Protein fusion: in frame (NCOA4-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022818-T01-IM6-181	P-0022818-T01-IM6	P-0022818	Colorectal Cancer	COAD	RET-NCOA4	NCOA4	unknown	three_prime	43611746	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup | Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-0009592-T01-IM5-182	P-0009592-T01-IM5	P-0009592	Thyroid Cancer	THPA	RET-NCOA4	NCOA4	unknown	three_prime	43611143	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET exons 12=20: c.872:NCOA4_c.2136+621:RETdup | Protein fusion: mid-exon (NCOA4-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002145-T01-IM3-183	P-0002145-T01-IM3	P-0002145	Thyroid Cancer	THPD	RET-PDCD10	PDCD10	out-of-frame	five_prime	43609394	10	627	True	626	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020975) - PDCD10 (NM_145860) translocation: t(10,3)(q11.21;q26.1)(chr10:g.43609394::chr3:g.167419184) | Protein fusion: out of frame (RET-PDCD10)	NA
EVT-P-0001898-T01-IM3-184	P-0001898-T01-IM3	P-0001898	Non-Small Cell Lung Cancer	LUAD	RET-RASSF4	RASSF4	in-frame	five_prime	43611749	11	712	True	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020975) - RASSF4 (NM_032023) deletion: c.2137-283_c.138+357del | Protein fusion: in frame (RET-RASSF4)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030396-T02-IM6-185	P-0030396-T02-IM6	P-0030396	Non-Small Cell Lung Cancer	LUAD	RET-RBPMS	RBPMS	unknown	three_prime	43610018	11	657	False	657	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RBPMS (NM_001008712) - RET (NM_020975) Rearrangement : t(8,10)(p21.1,q11.21)(chr8:g.30412656::chr10:g.43610018) | Protein Fusion: mid-exon {RBPMS:RET}	NA
EVT-P-0029222-T02-IM6-212	P-0029222-T02-IM6	P-0029222	Thyroid Cancer	THPA	RET-RUFY1	RUFY1	unknown	three_prime	43610037	11	663	False	663	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RUFY1 (NM_025158) - RET (NM_020975) Rearrangement : t(5,10)(q35.3,q11.21)(chr5:g.179025890::chr10:g.43610037) | Protein Fusion: mid-exon {RUFY1:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015310-T02-IM6-213	P-0015310-T02-IM6	P-0015310	Thyroid Cancer	THPA	RET-RUFY3	RUFY3	out-of-frame	three_prime	43610427	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655316::chr10:g.43610427) | Protein fusion: out of frame (RUFY3-RET)	NA
EVT-P-0015310-T04-IM6-214	P-0015310-T04-IM6	P-0015310	Thyroid Cancer	THPA	RET-RUFY3	RUFY3	unknown	three_prime	43610427	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655251::chr10:g.43610427) | Protein Fusion: mid-exon {RUFY3:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025609-T01-IM6-215	P-0025609-T01-IM6	P-0025609	Esophagogastric Cancer	STAD	RET-SHROOM3	SHROOM3	in-frame	three_prime	43611545	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		SHROOM3 (NM_020859) - RET (NM_020975) rearrangement: t(4;10)(q21.1;q11.21)(chr4:g.77664882::chr10:g.43611545) | Protein Fusion: in frame {SHROOM3:RET}	NA
EVT-P-0053530-T01-IM6-216	P-0053530-T01-IM6	P-0053530	Colorectal Cancer	COAD	RET-SLC4A4	SLC4A4	out-of-frame	five_prime	43610439	11	712	True	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		RET (NM_020975) - SLC4A4 (NM_001134742) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.72412310::chr10:g.43610439) | Protein Fusion: out of frame {RET:SLC4A4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002145-T01-IM3-217	P-0002145-T01-IM3	P-0002145	Thyroid Cancer	THPD	RET-TFG	TFG	in-frame	three_prime	43609437	11	627	True	627	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RET (NM_020975) - TFG (NM_001195478) translocation: t(10,3)(q11.21;q12.2)(chr10:g.43609437::chr3:g.100450751) | Protein fusion: in frame (TFG-RET)	NA
EVT-P-0017716-T01-IM6-218	P-0017716-T01-IM6	P-0017716	Thyroid Cancer	THPA	RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43611172	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-7 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021980-T01-IM6-219	P-0021980-T01-IM6	P-0021980	Non-Small Cell Lung Cancer	LUAD	RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43610550	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused to RET exons 12- 20): c.763-591:NCOA4_c.2136+366:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0025531-T01-IM6-220	P-0025531-T01-IM6	P-0025531	Non-Small Cell Lung Cancer	LUAD	RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43611375	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NCOA4 (NM_001145260) RET (NM_020975) fusion: c.619-108 NCOA4_c.2137-657RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034704-T01-IM6-222	P-0034704-T01-IM6	P-0034704	Thyroid Cancer	THPA	RET-TRIM24	TRIM24	in-frame	three_prime	43611827	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		TRIM24 (NM_015905) - RET (NM_020975) Fusion (TRIM24 exon 9 fused with RET exon12) : t(7;10)(q34;q11.21)(chr7:g.138243011::chr10:g.43611827) | Protein Fusion: in frame {TRIM24:RET}	NA
EVT-P-0046790-T01-IM6-223	P-0046790-T01-IM6	P-0046790	Hepatobiliary Cancer	GBC	RET-UXS1	UXS1	unknown	five_prime	43600430	4	219	False	218	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain	Cadherin domain	RET (NM_020975) - UXS1 (NM_001253875) rearrangement: t(2;10)(q12.2;q11.21)(chr2:g.106728260::chr10:g.43600430) | Protein Fusion: mid-exon {RET:UXS1}	NA

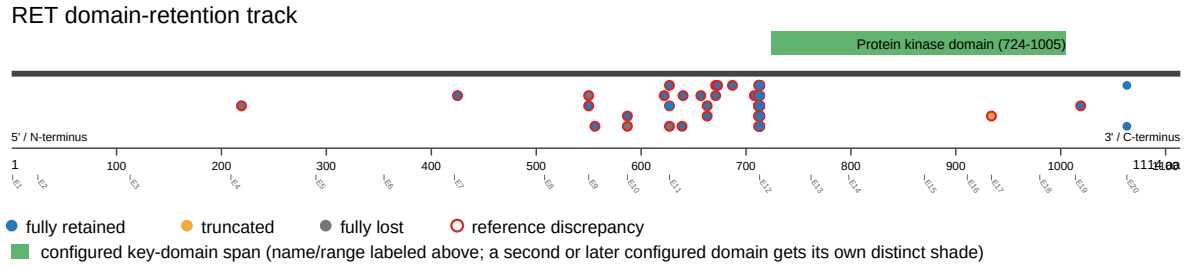
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029222-T03-IM6-224	P-0029222-T03-IM6	P-0029222	Thyroid Cancer	THPA	RUFY1-RET	RUFY1	unknown	three_prime	43610037	11	663	False	663	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RUFY1 (NM_025158) - RET (NM_020975) fusion: t(5;10)(q35.3;q11.21)(chr5:g.179025890::chr10:g.43610037) | Protein Fusion: mid-exon {RUFY1:RET}	NA
EVT-P-0031340-T02-IM6-225	P-0031340-T02-IM6	P-0031340	Non-Small Cell Lung Cancer	LUAD	RUFY2-RET	RUFY2	in-frame	three_prime	43611559	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		RUFY2 (NM_017987) - RET (NM_020975) fusion: c.1045-640:RUFY2_c.2137-473:RETinv | Protein Fusion: in frame {RUFY2:RET}	NA
EVT-P-0036400-T06-IM6-227	P-0036400-T06-IM6	P-0036400	Renal Cell Carcinoma	SRCC	SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		SPECC1L (NM_001145468) - RET (NM_020975) fusion: t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) | Protein Fusion: in frame {SPECC1L:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026614-T01-IM6-228	P-0026614-T01-IM6	P-0026614	Breast Cancer	ILC	SPECC1L-RET	SPECC1L	in-frame	three_prime	43611575	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-10 fused to RET exons 12-20): t(10;22)(10q11.21;22q11.23)(chr10:g.43611575::chr22:g.24740413) | Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-0036400-T03-IM6-229	P-0036400-T03-IM6	P-0036400	Soft Tissue Sarcoma	SARCNOS	SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) | Protein Fusion: in frame {SPECC1L:RET}	NA

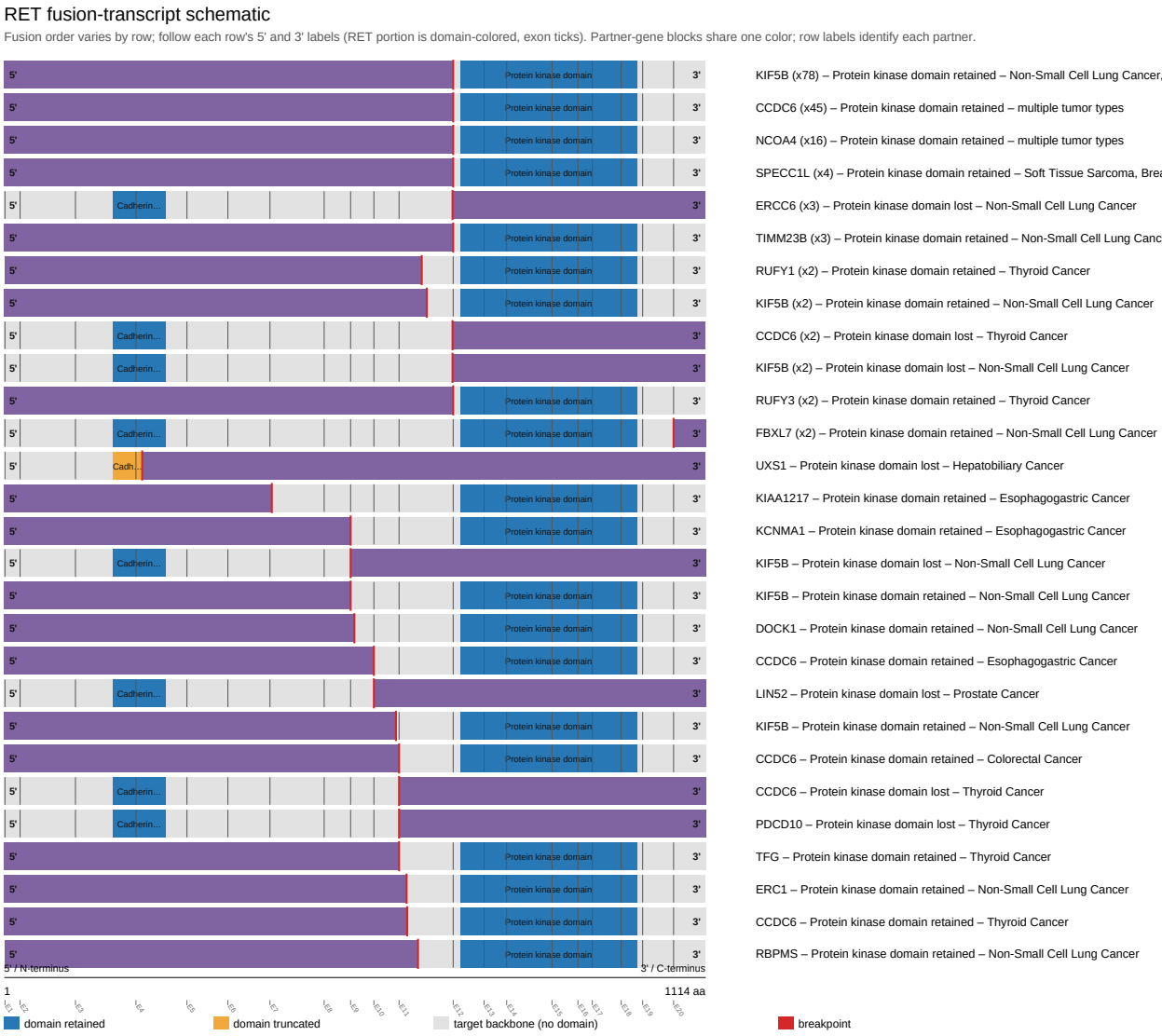
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036400-T04-IM6-230	P-0036400-T04-IM6	P-0036400	Soft Tissue Sarcoma	SARCNOS	SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	12	713	True	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) | Protein Fusion: in frame {SPECC1L:RET}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 45 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

